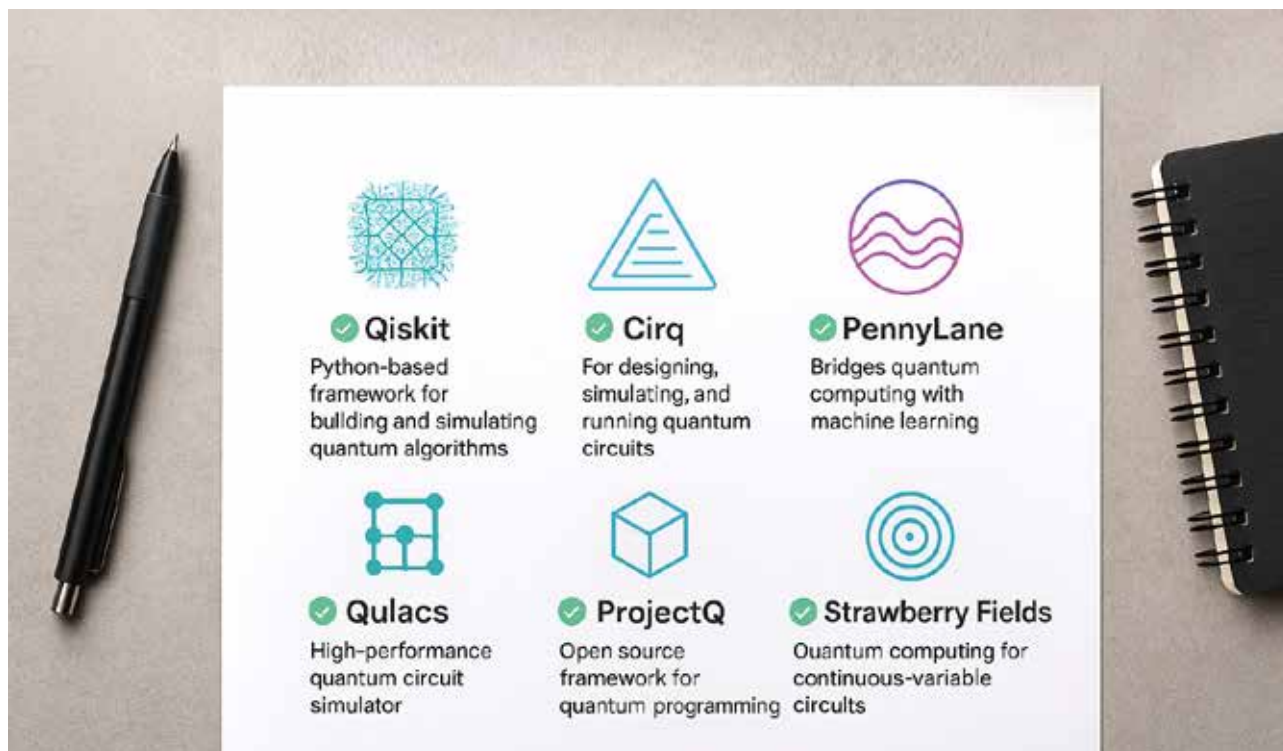


The Top Open Source Quantum Computing Frameworks

Open source could well be the backbone of quantum computing. Explore how the most popular open source quantum computing frameworks work and what they can be used for.



Quantum programming differs significantly from conventional programming. A developer or researcher must deal with abstract concepts such as superposition, entangled particles, and noisy qubit (quantum bit) interactions in quantum computing, making it difficult to work without the proper tools. The good news is that there are open source frameworks available today that allow developers to prototype or theorise a quantum algorithm in either simulation environments or real hardware.

In classical computing, information is stored as bits, where each bit can have the value of either 0 or 1. In quantum computing, the basic unit of information is a qubit, which can have multiple values 'at once' (superposition) and can relate across states (entangled).

As programming at the physical qubit level is complicated, software frameworks will be necessary. Frameworks provide abstractions for constructing circuits, include various libraries of algorithms (like those used in classical computing frameworks), and provide some interface for simulators or

real quantum devices.

Open source frameworks are playing a key role in lowering the barriers of quantum software development as different stakeholders can potentially contribute to them (academics, startups and industry). These frameworks offer transparency and affordability, and enable rapid prototyping.

While most of the leading frameworks used today are maintained by the leading tech companies (IBM, Google, D-Wave, Xanadu, and Rigetti) they are still free and available to the community.